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Re: Lead Robotics Safety Engineer

I've spent the last five years building functional safety for Vay, a teledriving company, in a domain that had no established regulation when I started — close to where robotics is now, with technology, standards, and regulation all evolving at once. I'm applying for the Lead Robotics Safety Engineer role because that exact pattern, building safety into the unknown rather than waiting for the rules to catch up, is the work I want to keep doing.

When I joined Vay, there was no regulatory framework anywhere for remotely operated vehicles, and I enjoyed building into that ambiguity rather than waiting for it to resolve. The safety director and I built the in-house safety function from scratch and grew it as the company scaled, deriving its safety framework from first principles: system-level hazard analysis, risk assessment, safety cases, and a methodology for deciding what level of risk is tolerable in the field. One concrete output of that work is a set of tolerable failure-rate thresholds, derived from probabilistic risk analysis, for each type of safety fallback our vehicles can trigger; if the observed rate of any one exceeds its threshold, it automatically triggers a stop-fleet decision. That framework was tested by an independent TÜV SÜD safety assessment, whose positive endorsement made Vay the first company in Europe permitted to operate on public roads with no driver in the vehicle. The fleet now runs at roughly 200 vehicles in Las Vegas, with expansion underway to other US cities, and the same safety case and risk framework have scaled with it. I'm now leading the safety work as we bring AI into the driving task itself.

Day to day, this work means partnering with engineering, leadership, and legal so that safety is part of the architecture from the start rather than a late check, monitoring and applying regulation as it develops, and representing the safety case to an external assessor. I recognise that your preferred standards (ISO 10218, ISO 13849, IEC 61508, ISO 12100, ANSI/RIA) sit in robotics and machinery rather than automotive, and that the standards and certification-body experience I bring is from an adjacent domain. What I'd bring directly is the harder-to-teach part: the discipline of building and scaling a safety function, and a tolerable-risk framework with real operational teeth, from scratch, before the rules existed to tell me how.

I'd welcome the chance to talk about how building a safety function from first principles, before the rules existed, applies to building and scaling one in robotics.

Best Regards,

Sourabh Porwal